Quantum Mechanics of Electrical Current Circuits

An informal lecture

Dr. Oleg Starykh
Department of Physics
University of Utah

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Outline

- Basic Quantum Mechanics: particle-wave duality
- Quantum wires: man-made and natural
- Q.M. + Ohm = quantum wire: resistance is quantized!
- Tunneling into carbon nanotubes: violation of Ohm's law
- Charge density waves

Basic Quantum Mechanics

Max Planck (1900) - wave motion (thermal radiation, light) is quantized

$$E = n\,h\,f\ ,\ n = 0,1,2,3,\ldots$$

Planck's constant $h = 6.626\ .\ 10^{-34}$ Joule.sec

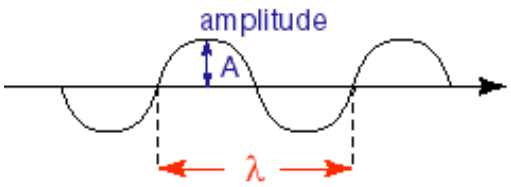
Frequency = f , period T = 1/f

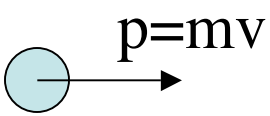
Albert Einstein (1905) - light (frequency f) is a particle (photon)

$$E = h\,f$$

Louis de Broglie (1924) - matter is wave: $E=hf$, $p = h/\lambda$

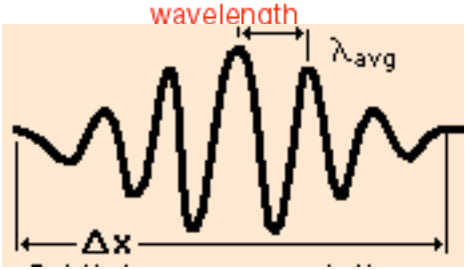
wavelength = λ



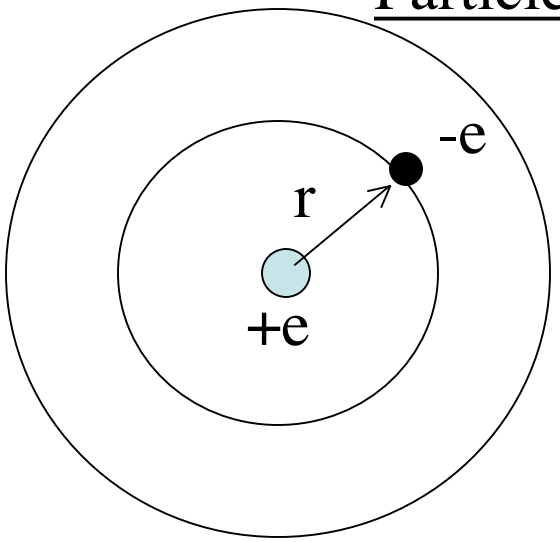


?

$$\vec{p} = m\vec{v}, E = \frac{mv^2}{2} = \frac{p^2}{2m}$$

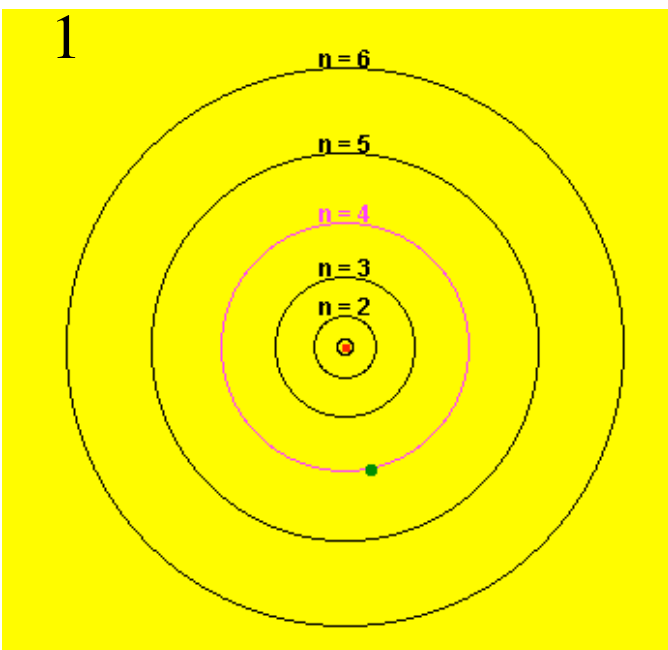


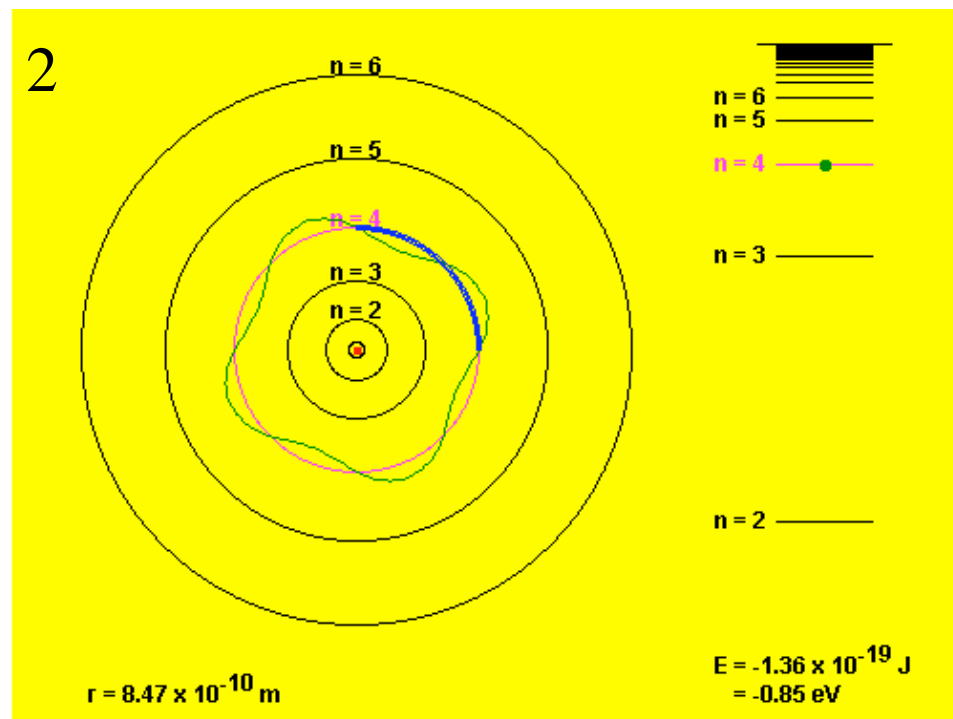
Particle-wave duality: Bohr's atom (1913)



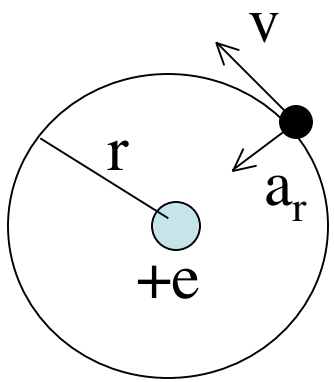
1. -e moves in a circular orbit (like a planet around the Sun)

2. Only certain quantized orbits are allowed: those with circumference = multiple of (-e) wavelength λ: $2\pi r = n\,\lambda$ (standing wave forms the orbit!)





A bit of algebra



From (1): $$\frac{k_e e^2}{r^2} = m a_r = m\frac{v^2}{r} \rightarrow k_e e^2 = m v^2 r$$

(Newton's 2nd law for Coulomb force)

Orbital angular momentum is conserved:
L=mv r = pr = const

From (2): $2\pi r = n\lambda \Rightarrow r = n\,\lambda/(2\pi)$ and $p = h/\lambda$.
Thus $L = p\,r = (h/\lambda)\,(n\lambda/2\pi) = n\,h/2\pi$ is quantized! $\hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34} J \cdot s$
n=1,2,3,…is integer

Find - speed and radius are quantized:

$$v_n = \frac{k_e e^2}{\hbar n}, v_{n=1} = 2 \times 10^6 m/s$$

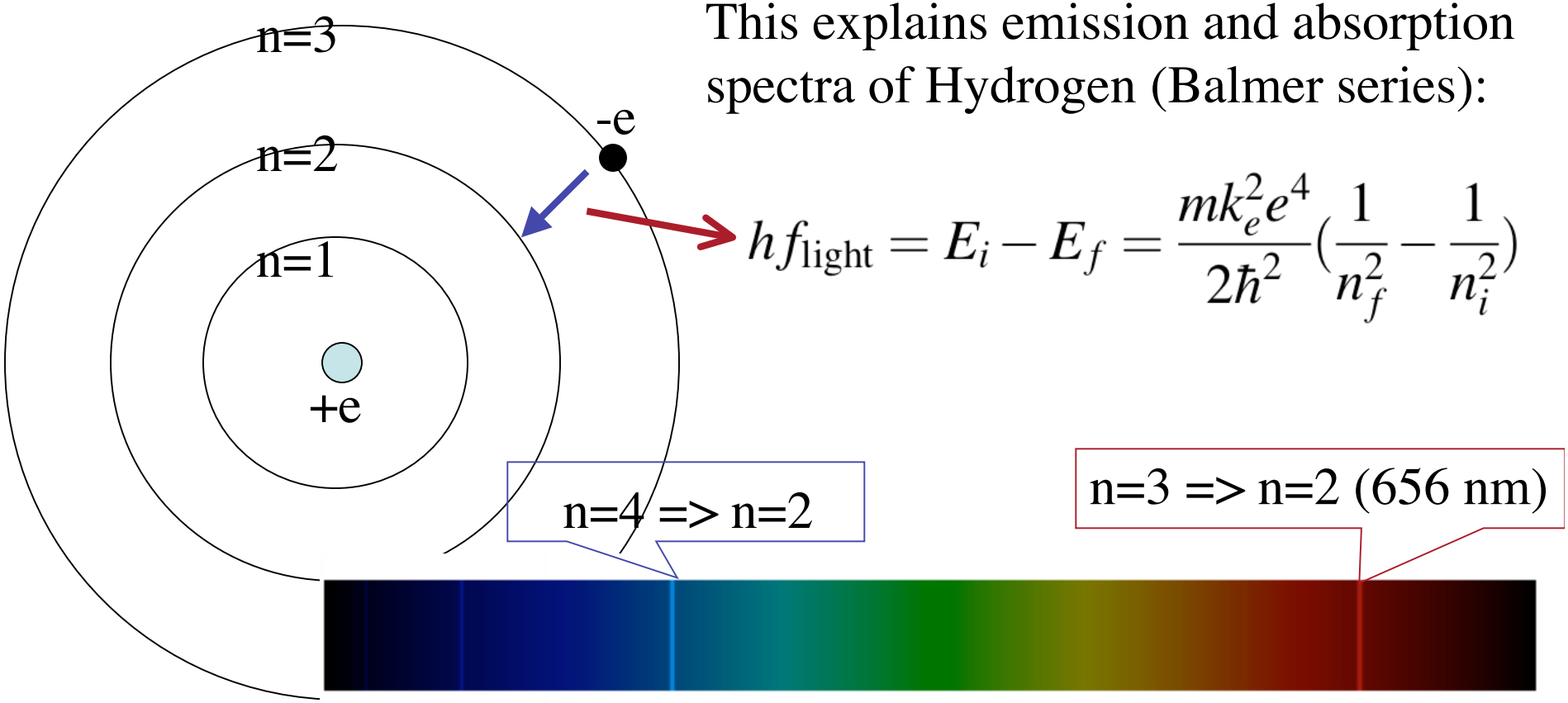
$$r_n = \frac{\hbar n}{m v_n} = \frac{\hbar^2 n^2}{m k_e e^2},\ r_{n=1} \approx 0.5 \times 10^{-10}\ m = 0.5 Å$$

And energy is quantized as well - it's quantum mechanics!

$$E_n = \frac{mv_n^2}{2} - \frac{k_e e^2}{r_n} = -\frac{mk_e^2 e^4}{2\hbar^2}\frac{1}{n^2}$$

This explains emission and absorption spectra of Hydrogen (Balmer series):

$$hf_{\text{light}} = E_i - E_f = \frac{mk_e^2 e^4}{2\hbar^2}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$



Semiconductor quantum point contact

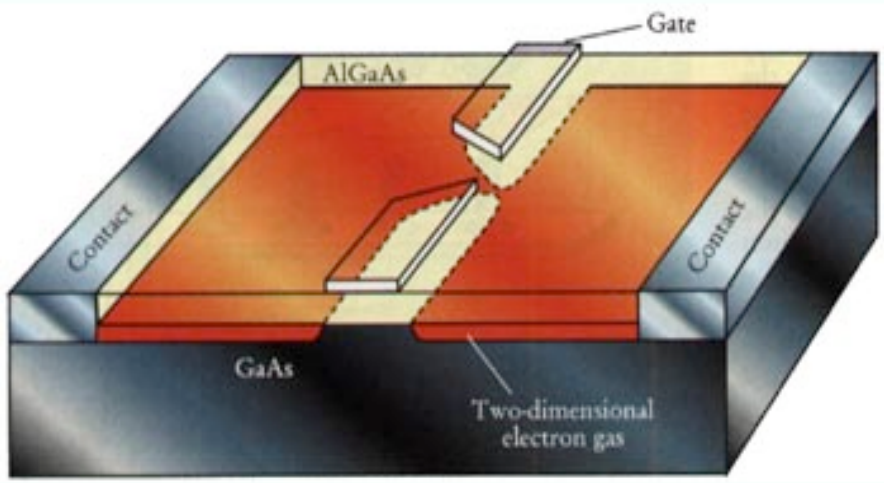


Figure 1. *Schematic cross-sectional view of a quantum point contact, defined in a high-mobility 2D electron gas at the interface of a GaAs-AlGaAs heterojunction. The point contact is formed when a negative voltage is applied to the gate electrodes on top of the AlGaAs layer. Transport measurements are made by employing contacts to the 2D electron gas at either side of the constriction.*

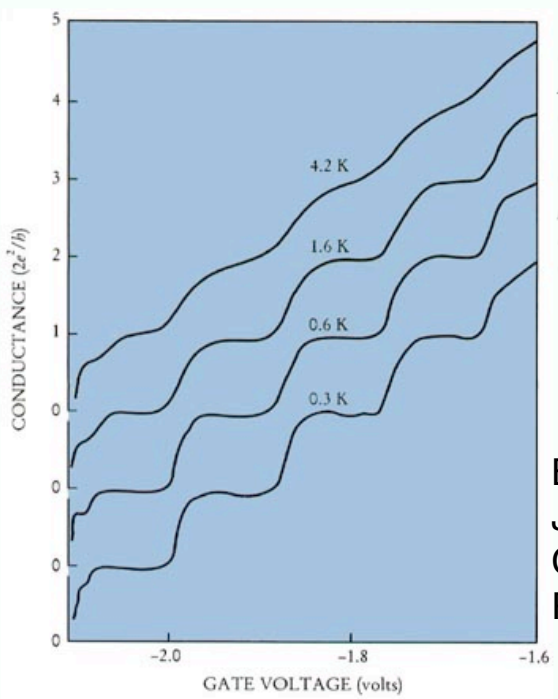


Figure 2. *Conductance quantization of a quantum point contact in units of $2e^2/h$. As the gate voltage defining the constriction is made less negative, the width of the point contact increases continuously, but the number of propagating modes at the Fermi level increases stepwise. The resulting conductance steps are smeared out when the thermal energy becomes comparable to the energy separation of the modes. (Adapted from ref. [6].)*

Cross-sectional area
is controlled
by the gate voltage.

B. J. van Wees, H. van Houten, C. W. J. Beenakker, J. G. Williamson, L. P. Kouwenhoven, D. van der Marel, C. T. Foxon, Phys. Rev. Lett. **60**, 848 (1988); Phys. Rev. B **43**, 12431 (1991).

Semiconductor quantum wire

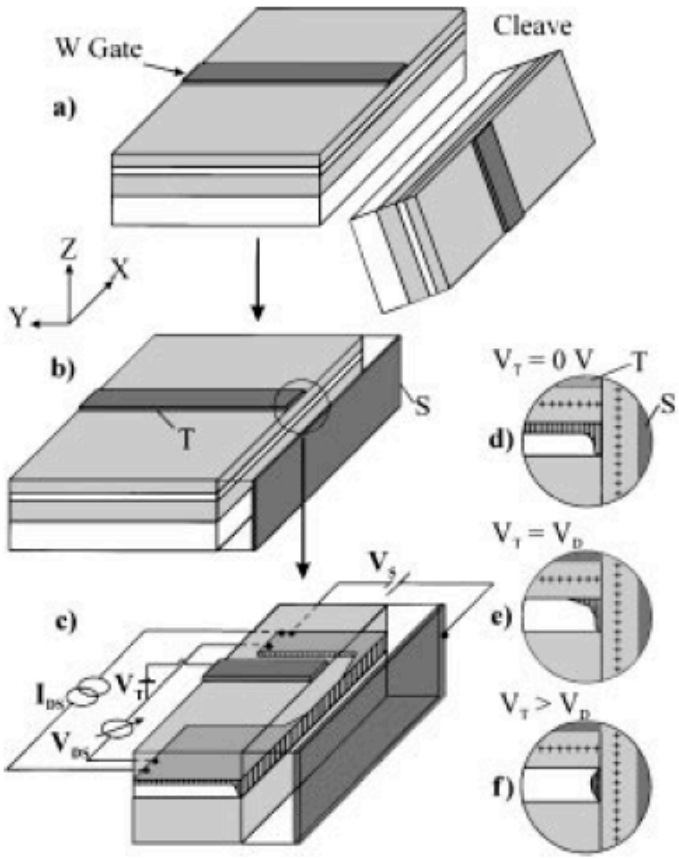


FIG. 1. (a) Wire preparation by cleaved edge overgrowth of GaAs-AlGaAs by molecular-beam epitaxy. For details see text.

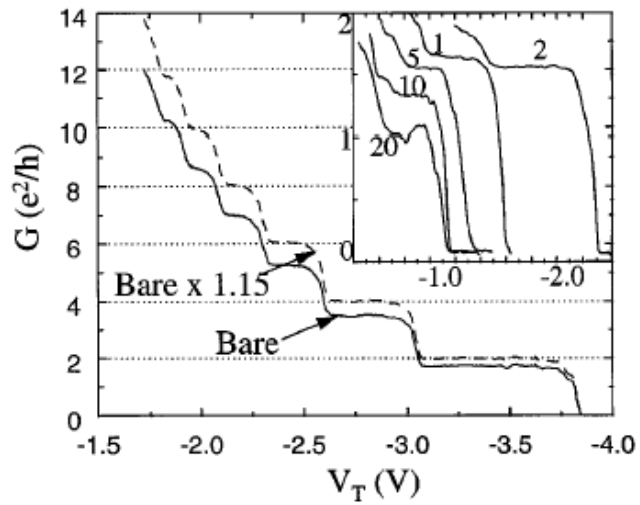


FIG. 2. Linear response conductance of a 2 μm long wire in a 25 nm quantum well vs the top-gate voltage (V_T) measured at a temperature of 0.3 K. The solid line is the measured conductance. The dashed curve is the measured conductance multiplied by an empirical factor of 1.15. Inset: Linear response conductance of the last plateau for wires of different lengths fabricated consecutively along the edge of a single 25 nm cleaved edge overgrowth specimen. The numbers denote the wire length in microns.

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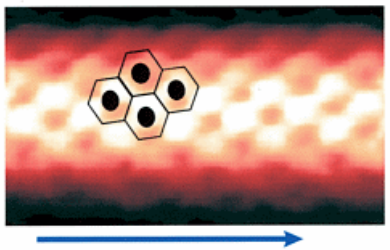
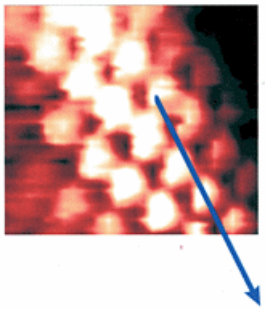
Nonuniversal Conductance Quantization in Quantum Wires

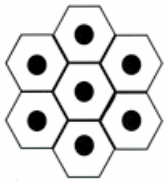
A. Yacoby,[1] H. L. Stormer,[1] Ned S. Wingreen,[2] L. N. Pfeiffer,[1] K. W. Baldwin,[1] and K. W. West[1]
[1]*Bell Laboratories, Lucent Technologies, Murray Hill, New Jersey 07974*
[2]*NEC Research Institute, Princeton, New Jersey 08540*

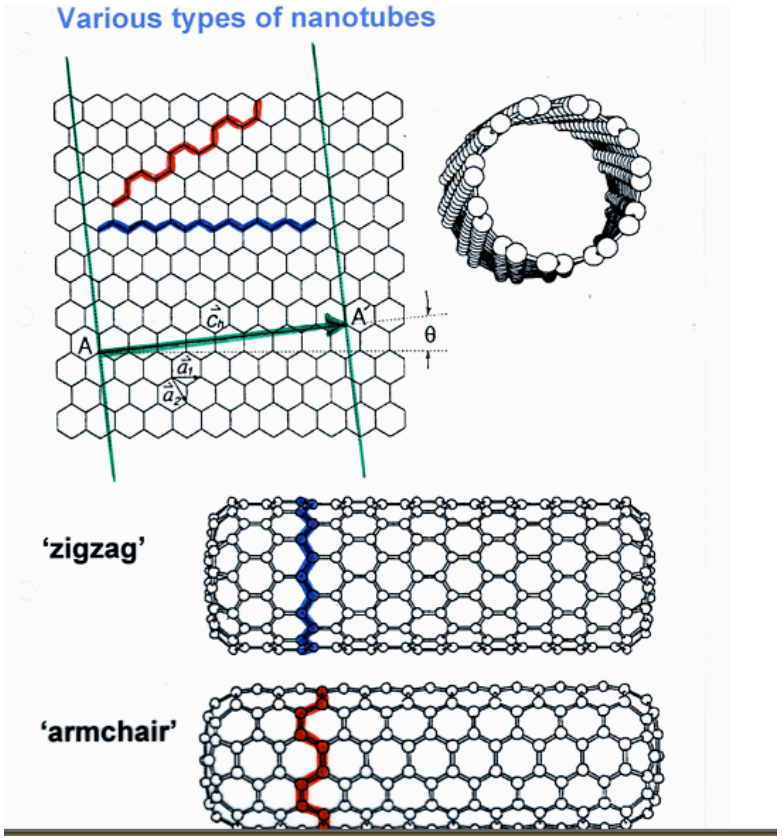
Carbon Nanotubes (discovered by S. Ijiima in 1991)

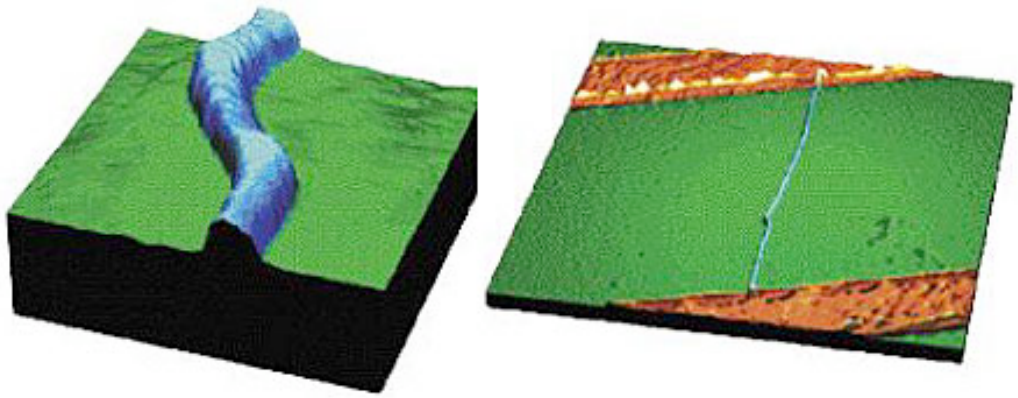




triangular lattice:
dark dots correspond
to centers hexagons



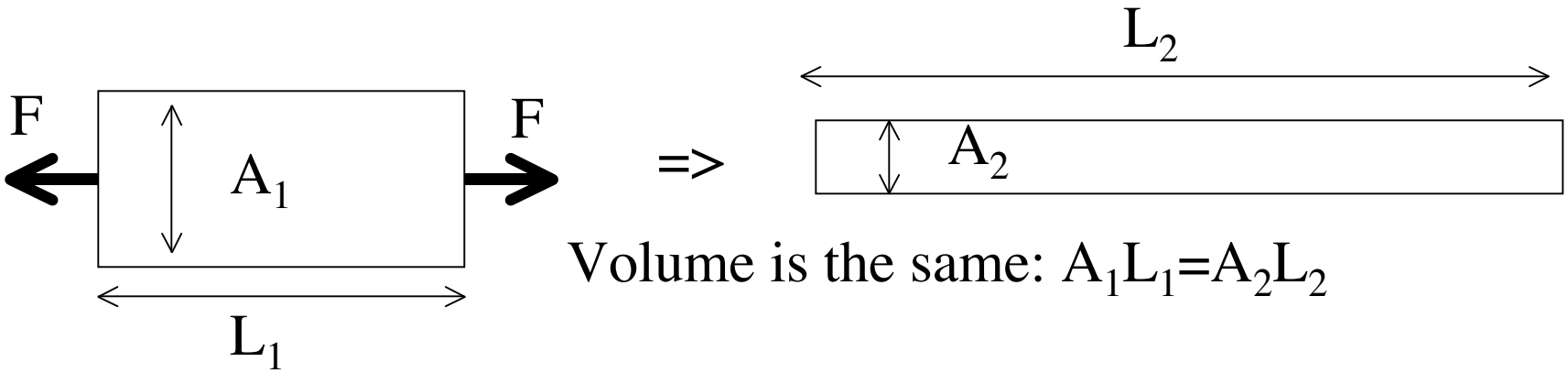




PHOTOS COURTESY OF CEES DEKKER AND HENK POSTMA/DELFT UNIVERSITY OF TECHNOLOGY

Cees Dekker,
Delft Univ. of Technology

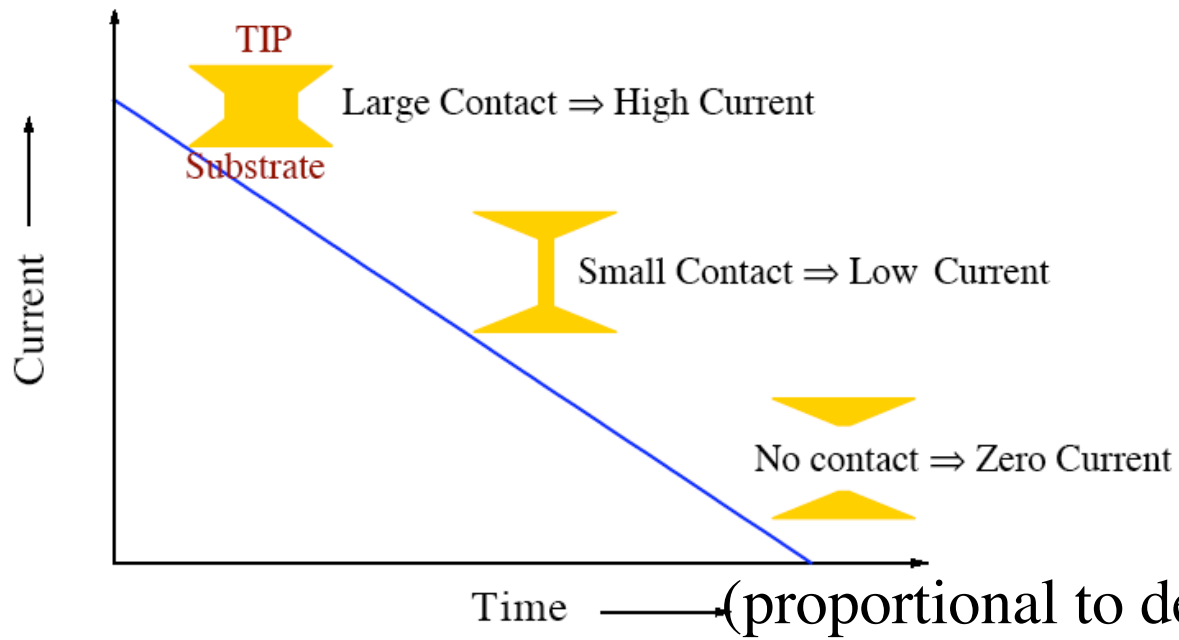
Break-junction: let's pull the wire!



Volume is the same: $A_1L_1=A_2L_2$

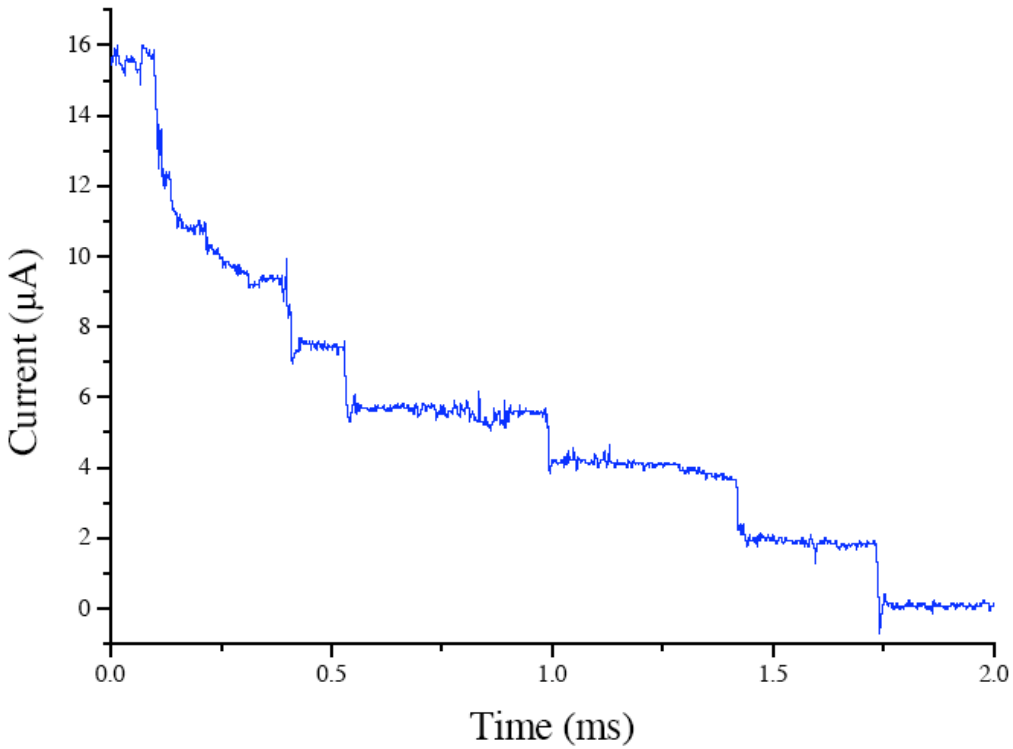
$G=ne^2t$ (Volume)/[m L^2] => $G_1 > G_2$ because $L_1 < L_2$.
That is, the resistance will increase $R_1 < R_2$ and current decrease, $I_1 > I_2$

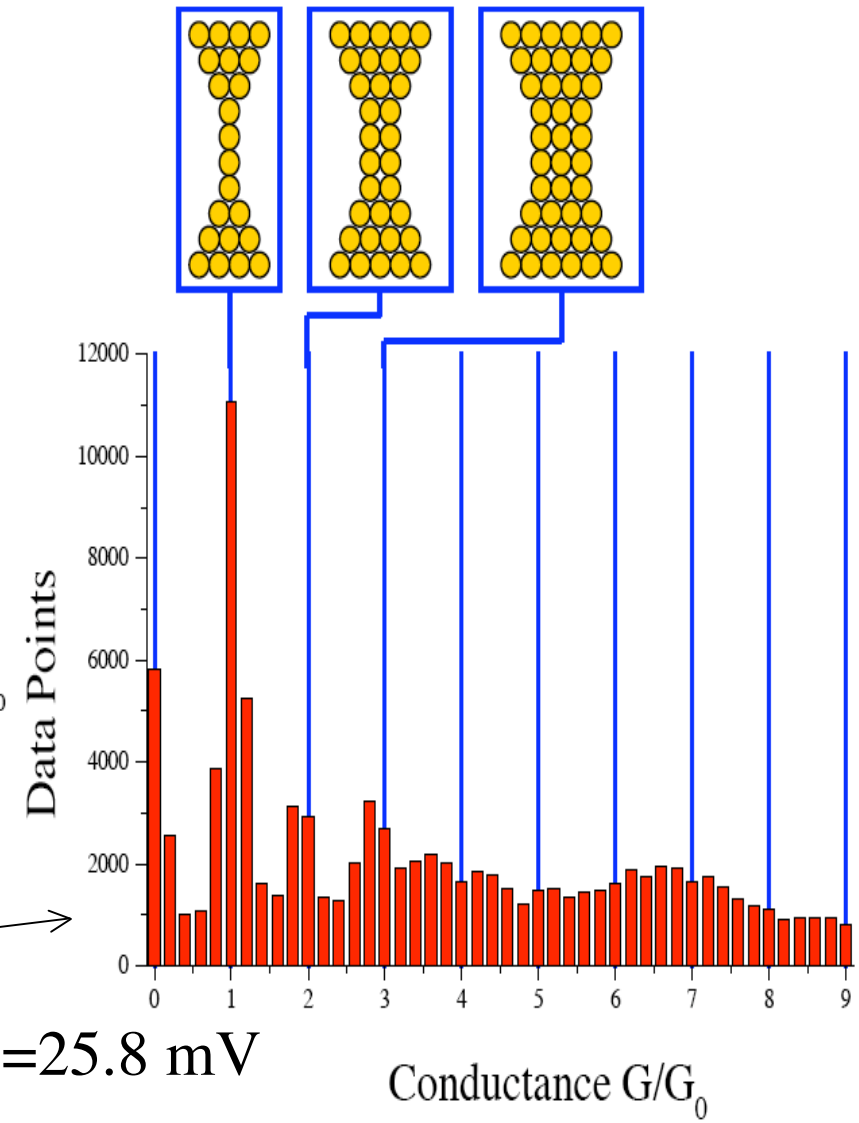
Expect:



(proportional to deformation)

Experimental data [LG. Soukiassian, R. Reifenberger, Purdue Univ.2000]





Step-wise decrease of the current!

Histogram of conductance G = I/V, V=25.8 mV

Conductance measurements

S. Franck et al., Science 280, 1744 (1998).

Complication - difficult to make good contacts.

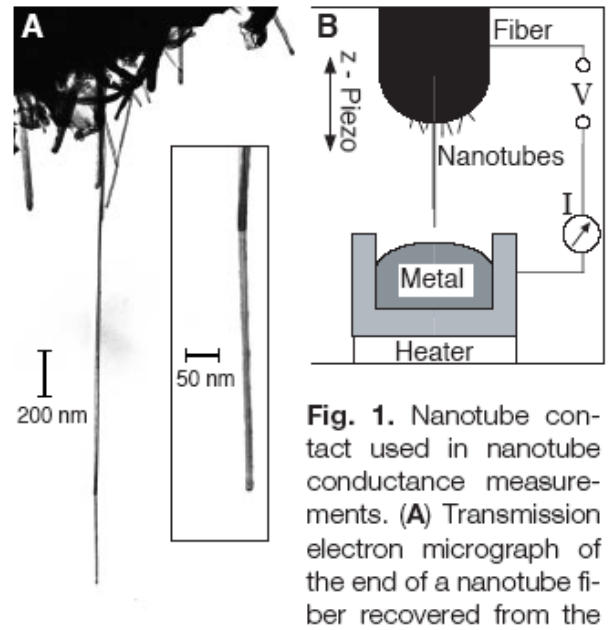
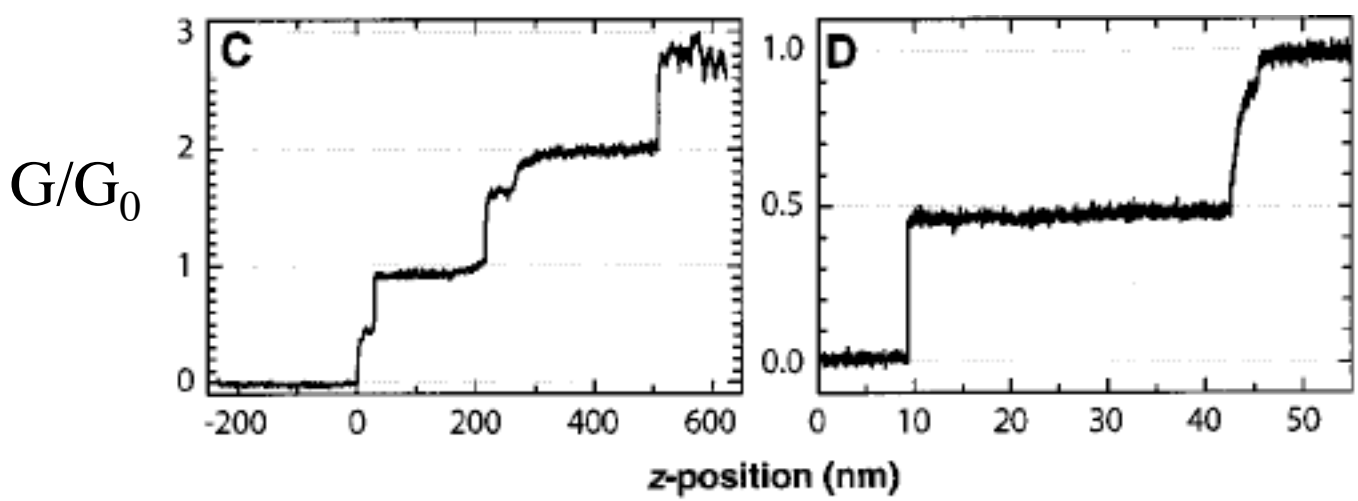
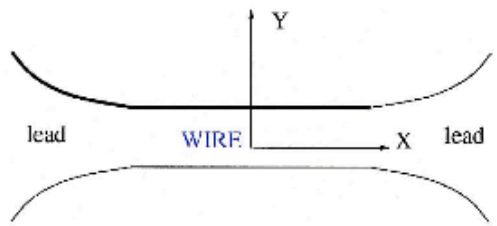
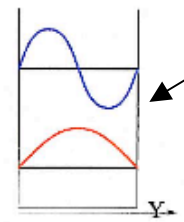


Fig. 1. Nanotube contact used in nanotube conductance measurements. (**A**) Transmission electron micrograph of the end of a nanotube fiber recovered from the



Quantum wire





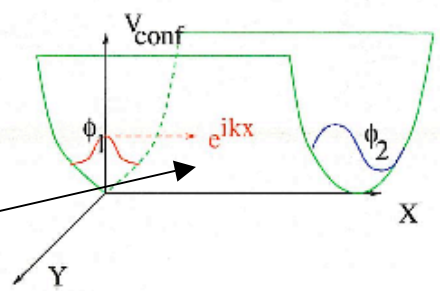
Standing waves in transverse (Y and Z) directions

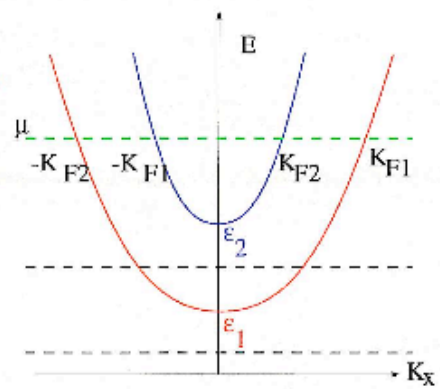
width $\propto \lambda_F \approx 300 - 400$ Å,
length $\sim 1 - 10\mu m \ll L_{elastic}, L_{inelastic}$

- electron is confined in Y, Z directions but free to move in X.

$$H_0(x,y) = -\frac{\nabla_x^2}{2m} - \mu - \frac{\nabla_y^2}{2m} + V_{conf}(y)$$

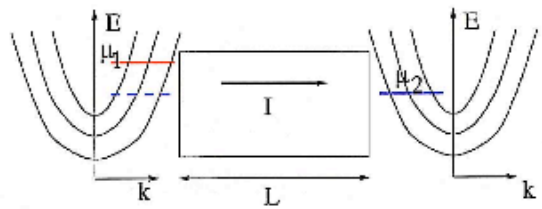
$$\Psi(x,y) = \sum_n \varphi_n(y)\ e^{\pm ik_n x} \Rightarrow E = \frac{k_n^2}{2m} - \mu + \epsilon_n$$





Plane wave along the wire (X axis)

Ballistic conductor



$$
\begin{aligned}
I_i &= e/t_i = e/(L/v_i), \quad v_i = \frac{1}{\hbar}\frac{\partial E_i}{\partial k} \\
I = \; & 2\sum_i e v_i/L = \frac{2e}{L}\sum_i \sum_k \frac{\partial E_i}{\hbar \partial k} \\
&= \frac{2e}{L}\sum_i \frac{L}{2\pi}\int dk \frac{\partial E_i}{\hbar \partial k}(f(E_i-\mu_1)-f(E_i-\mu_2)) \\
&= \frac{2e}{2\pi\hbar}\int dE M(E)(f(E-\mu_1)-f(E-\mu_2)) \\
&= \frac{2e^2}{h} M \frac{\mu_1-\mu_2}{e}
\end{aligned}
$$

$\Rightarrow I = GV, \quad G = \frac{2e^2}{h}M \Leftrightarrow I = V/R, \quad R = \frac{h}{2e^2}\frac{1}{M}$

Quantum of resistance $R_0 = G_0^{-1} = \frac{h}{2e^2} = 12,906\Omega$

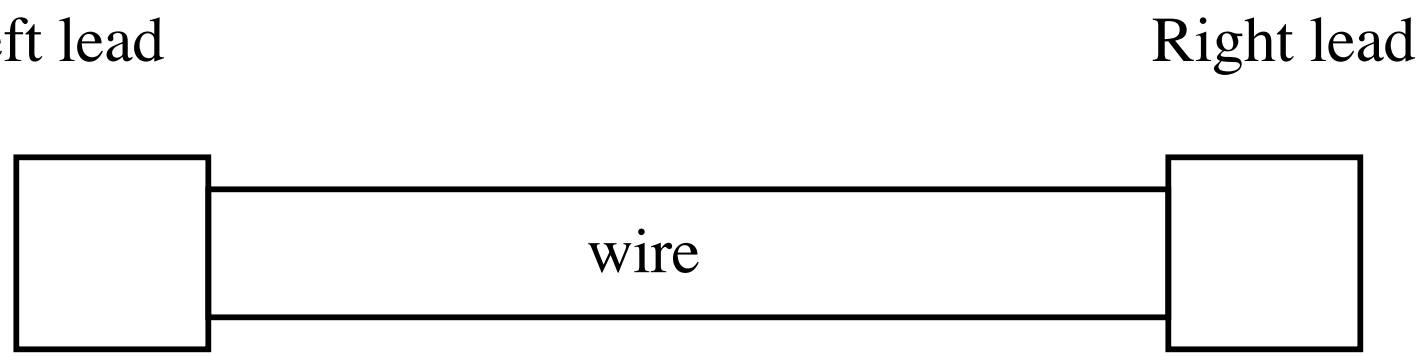
index i = subband

M: # subbands

spin

degeneracy

Fermi distribution function



Electrons moving to the Right

$$\int_0^{\mu_1} dE\, f(E-\mu_1)$$

Electrons moving to the Left

$$\int_0^{\mu_2} dE\, f(E-\mu_2)$$

The difference at T=0: $\mu_1 - \mu_2 = e\,V$

V = potential difference

Thus the current $I = (2e^2/h)\,V$ for one conducting channel

Hence: conductance $G = 2*(e^2/h)*$(# conducting channels/transverse subbands)

Dimensional Analysis

What are the units of e^2/h ?

$$\left.\begin{array}{l}[e]=\text{charge}=\text{Coulomb}=C \\ [h]=\text{Planck's constant}=\text{Joule*sec}=J\ s\end{array}\right\} \quad [e^2/h]=C^2/(J\ s) = (C/s)\ (C/J)$$

Current = Charge per second = C/s = Ampere = A

Electric potential = Voltage = Volts = V= Work per unit charge = J/C

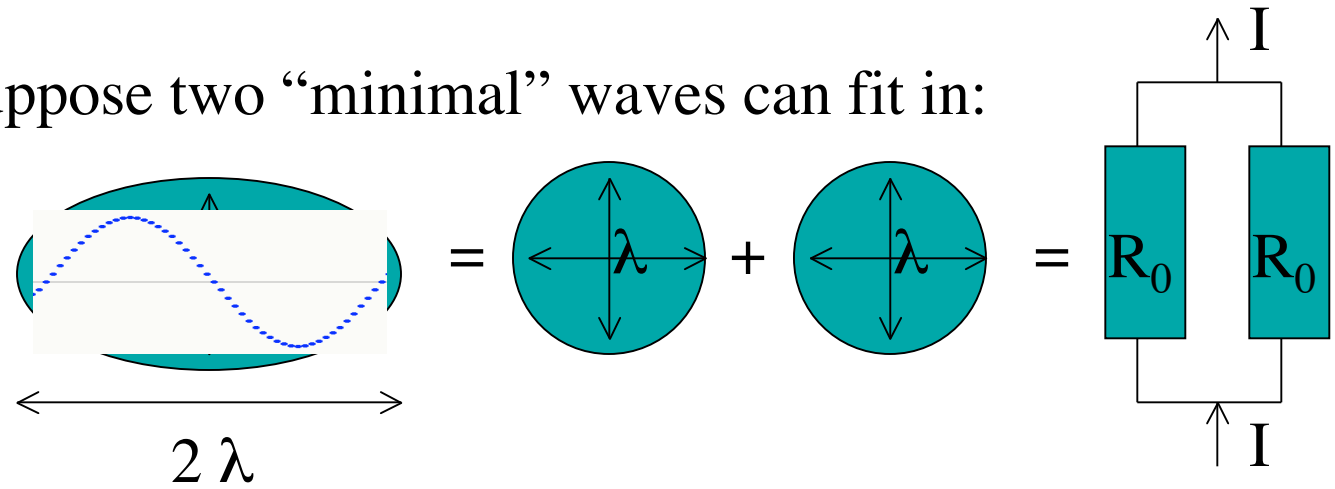
Thus, C/J = 1/V, and $[e^2/h]$ = A/V

I = V/R => Resistance = R = V/I = Volt/Ampere = Ohm = Ω

Finally, $[e^2/h]$ = A/V = 1/Ω = 1/Ohm = 1/(Resistance) = Conductance

Go backward: increase the cross-sectional area

Suppose two "minimal" waves can fit in:



Two "maximal" resistors in parallel: $I = 2\ V/R_0$ => $R=R_0/2=h/(4e^2)$

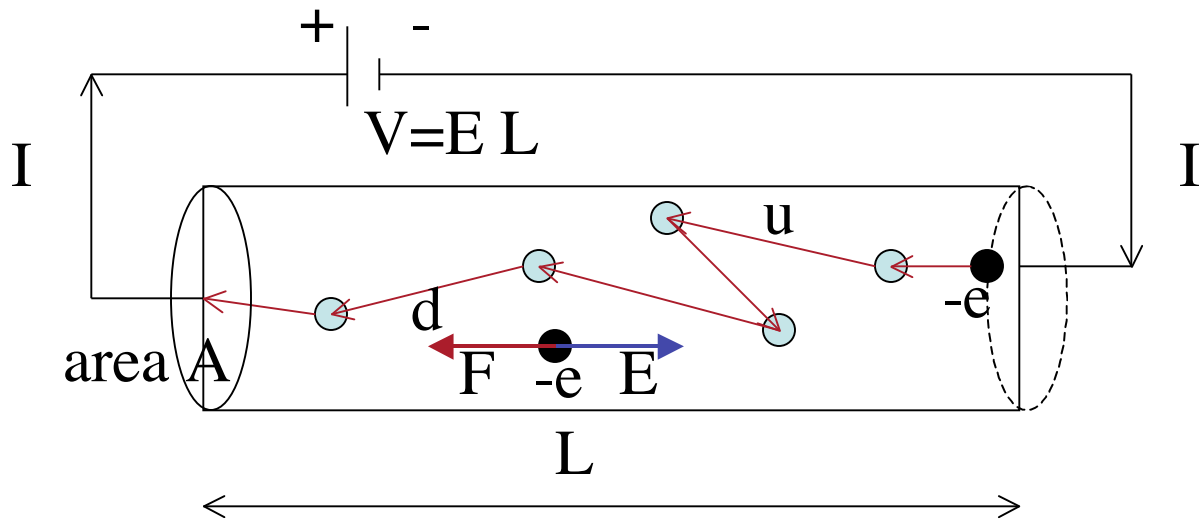
Hence, double conductance $G= 2G_0=2\ (2e^2/h) = 4e^2/h$

Thus expect $G = N\ G_0 = N\ (2e^2/h)$: step-wise increase of conductance as the number of conducting channels increases with cross-section A, N=1,2,3,…

<u>This is conductance quantization</u>

Bigger (macroscopic) wire: Ohm's law

$I = V/R = G\,V$ R=resistance, G=conductance



Electrons move with randomly oriented $u = 10^6$ m/s; d=mean-free path (average distance between impurities; 10^{-7}-10^{-6} m)

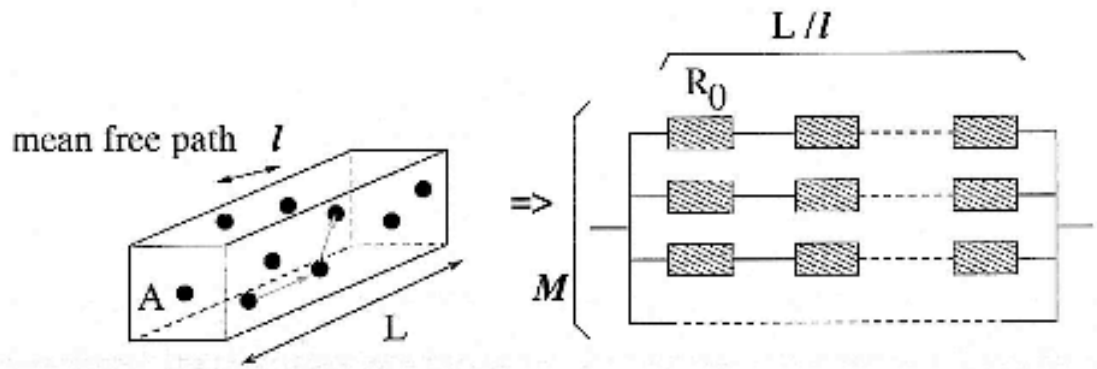
$F = e\,E$, $F = m\,a$ } $a = eE/m$, $t = d/u$ } Velocity change between two consecutive collisions (drift velocity) $v_d = at = eEt/m$ [for copper $v_d = 1$ m/s << $u = 10^6$ m/s]

Current $I = (n\,e\,v_d)\,A = ne^2tA/(mL)\,V = G\,V$, $G = 1/R = ne^2t\,A/(mL)$

(Drude formula)

n = number of electrons per unit volume (concentration)

Diffusive conductor



Ohm's law: $I = V/R, \quad R = \rho L/A, \quad G = \sigma A/L$

- Ohm vs Landauer:

· no quantum interference $\Rightarrow L/\ell$ resistors with $R_0 = \frac{h}{2e^2}$ in series $\Rightarrow R_s = \frac{h}{2e^2}\frac{L}{\ell}$

· $\mathcal{M} \sim A/\lambda_F^2 = Ak_F^2$ conducting channels $\Rightarrow$ total resistance $R_T = R_s/\mathcal{M} \sim \frac{h}{2e^2} L/(\ell k_F^2 A)$

· $\sigma = \rho^{-1} = \frac{e^2}{h} v_F \tau k_F^2 \sim \frac{e^2 k_F^3 \tau}{m}$ [using $v_F = \hbar k_F/m$]

· electron density $n \sim \int^{k_F} d^3\vec{k} \sim k_F^3$

Conclusions

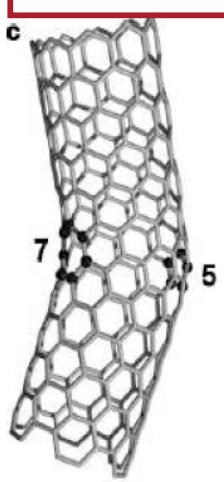
Quantum Mechanics governs atoms (size ~ 10^{-10} m) and nano-scale conductors (size ~ 10^{-8}-10^{-6} m).

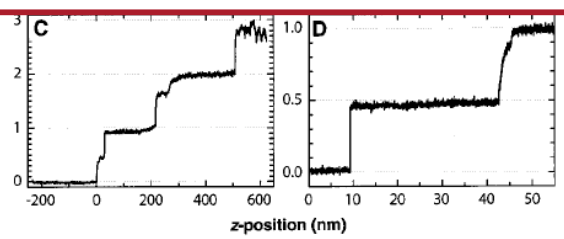
Electron is both particle and wave.

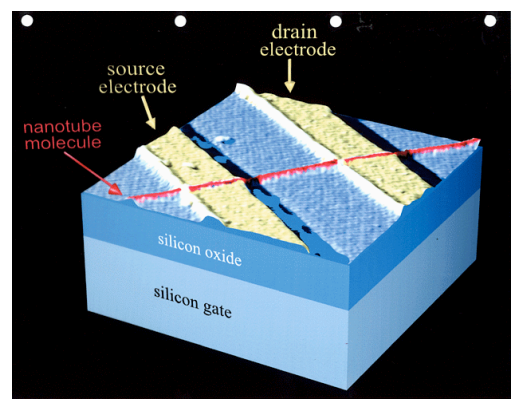
Particle-like: electrical current in macroscopic samples ($L > \lambda = h/p$).

Wave-like: current in nano-scale samples (quantum point contacts and quantum wires; size $L < \lambda$=wavelength).

Conductance (and resistance) is quantized: $G = N\,G_0$, $G_0 = 2e^2/h$
N=(number of channels)=0,1,2,3,… $R=1/G = R_0/N$, $R_0 = h/(2e^2)$





Nanotechnology:

Small is beautiful (and useful)